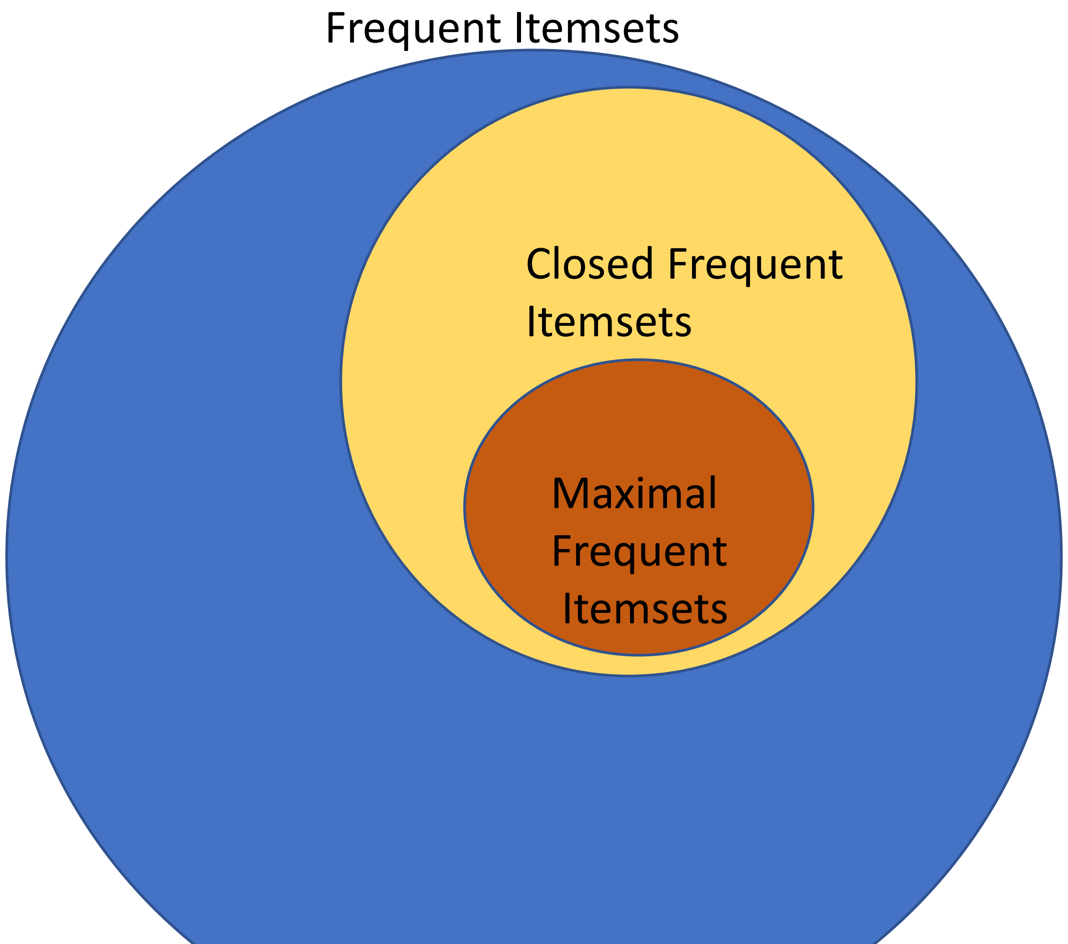
Frequent Itemsets, Closed Itemsets, Maximal Itemsets

Frequent itemset: A frequent itemset is an itemset whose support is greater than some specified minimum support.

Closed Frequent itemset: An itemset is closed if none of its immediate supersets has the same support as that of the itemset.

Maximal Frequent itemset: An itemset maximal frequent if none of its immediate supersets is frequent.

Relation between Frequent Itemsets, Closed Itemsets and Maximal Itemsets



TID	List of Items
T1	A,B,C,D
T2	A,B,C,D
T3	A,B,C
T4	B,C,D
T5	C,D

Min _Support =3

Item	Count
A	3
B	4
C	5
D	4

All items are frequent

Items	Count
A,B	3
A,C	3
A,D	2
B,C	4
B,D	3
C,D	4

Min _Support =3

To find A is closed , Maximal

A(3)
AB(3) , AC(3),AD(2)
A is not closed since all its immediate superset support count is not less than support count of A.

A is not maximal since its immediate superset are present with minimum support count.

To find B is closed or Maximal

B(4)
AB(3) , BC(4),BD(3)
B is not closed since all its immediate superset support count is not less than support count of B.

B is not maximal since its immediate superset are present with minimum support count.

To find C is closed or Maximal

C(5)
AC(3) , BC(4),CD(2)

C is closed since all its immediate superset support count is less than support count of C.

C is not maximal since its immediate superset are present with minimum support count.

To find D is closed or Maximal

D(4)
AD(2) , BD(3),CD(4)

D is not closed since all its immediate superset support count is not less than support count of D.

D is not maximal since its immediate superset are present with minimum support count.

Check for all 2 itemsets

To find AB is closed or Maximal

AB(3)
ABC(3) , ABD(2)

AB is not closed since all its immediate superset support count is not less than support count of AB.

AB is not maximal since its immediate superset are present with minimum support count.

Items	Count
A,B	3
A,C	3
A,D	2
B,C	4
B,D	3
C,D	4

Items	Count
A,B	3
A,C	3
A,D	2
B,C	4
B,D	3
C,D	4

Check for all 2 itemsets

To find AC is closed or Maximal

AC(3)
ABC(3) , ACD(2)

AC is not closed since all its immediate superset support count is not less than support count of AC.

AC is not maximal since its immediate superset are present with minimum support count.

AD is not frequent

Items	Count
A,B	3
A,C	3
A,D	2
B,C	4
B,D	3
C,D	4

Check for all 2 itemsets

To find BC is closed or Maximal
BC(4)
ABC(3) , BCD(3)

BC is closed since all its immediate superset support count is less than support count of BC.

BC is not maximal since its immediate superset are present with minimum support count

Check for all 2 itemsets

To find BD is closed or Maximal
BD(3)
ABD(2) , BCD(3)

BD is not closed since all its immediate superset support count is not less than support count of BD.

BD is not maximal since its immediate superset are present with minimum support count

Items	Count
A,B	3
A,C	3
A,D	2
B,C	4
B,D	3
C,D	4

Check for all 2 itemsets

To find CD is closed or Maximal
CD(4)
ACD(2) , BCD(3)

CD is closed since all its immediate superset support count is less than support count of CD.

CD is not maximal since its immediate superset are present with minimum support count

Items	Count
A,B	3
A,C	3
A,D	2
B,C	4
B,D	3
C,D	4

Check for all 3 itemsets

Items	Count
A,B,C	3
A,B,D	2
A,C,D	2
B,C,D	3

To find ABC is closed or Maximal
ABC(3)
ABCD(2)

ABC is closed since all its immediate superset support count is less than support count of CD.

ABC is not maximal since its immediate superset are present with minimum support count

Check for all 3 itemsets

Items	Count
A,B,C	3
A,B,D	2
A,C,D	2
B,C,D	3

To find BCD is closed or Maximal
BCD(3)
ABCD(2)

BCD is closed since all its immediate superset support count is less than the support count of BCD.

BCD is maximal since its immediate superset are not present with minimum support count

ABCD is not frequent

Item	Frequent	Closed	Maximal
A	YEs	NO	NO
B	YES	NO	NO
C	YES	YES	NO
D	YES	NO	NO

Item	Frequent	Closed	Maximal
AB	YES	NO	NO
AC	YES	NO	NO
AD	NO	NO	NO
BC	YES	YES	NO
BD	YES	NO	NO
CD	YES	YES	NO

Item	Frequent	Closed	Maximal
ABC	YES	YES	YES
ABD	NO	NO	NO
ACD	NO	NO	NO
BCD	YES	YES	YES

ABCD
Not Freq
Not closed
Not Maximal